Lab Assignment: Inter-Subnet Routing with 4 Hosts across 2 Subnets

Name: _____ Date: _____

Question 1: IP Addressing Table

The network has two /24 subnets. Each host uses the router interface in its own subnet as its default gateway.

Device	IP Address	Subnet	Default Gateway
Host A1	192.168.10.10/24	Subnet A	192.168.10.1
Host A2	192.168.10.11/24	Subnet A	192.168.10.1
Host B1	192.168.20.10/24	Subnet B	192.168.20.1
Host B2	192.168.20.11/24	Subnet B	192.168.20.1
Router eth1	192.168.10.1/24	Subnet A	Not applicable
Router eth2	192.168.20.1/24	Subnet B	Not applicable

Question 2: How do devices communicate within the same subnet and between different subnets?

Devices in the same subnet communicate directly using their IP and MAC addresses. They do not need a router. Devices in different subnets communicate through a router. The sending device sends the packet to its default gateway. The router forwards the packet to the correct subnet.

Question 3: What is the purpose of a default gateway?

A default gateway is the router address that a device uses to communicate with other networks. If the default gateway is missing or incorrect, devices may still communicate inside their own subnet, but they cannot properly communicate with other subnets.

Question 4: What router interfaces are required?

The router needs two network interfaces: eth1 with 192.168.10.1/24 for Subnet A, and eth2 with 192.168.20.1/24 for Subnet B. Both interfaces must be enabled and configured with the correct IP addresses.

Question 5: Why must IP forwarding be enabled?

IP forwarding allows the router to receive packets on one interface and forward them through another interface. It was enabled using the command:

```
sysctl -w net.ipv4.ip_forward=1
```

This allows communication between Subnet A and Subnet B.

Network Testing Results

The following ping tests were completed in the Containerlab network. Each test received all four packets with 0% packet loss.

Test	Result
Host A1 to Host A2 (192.168.10.10 to 192.168.10.11)	PASS — 4/4 received; 0% loss
Host A1 to Host B1 (192.168.10.10 to 192.168.20.10)	PASS — 4/4 received; 0% loss
Host B1 to Host B2 (192.168.20.10 to 192.168.20.11)	PASS — 4/4 received; 0% loss
Host B1 to Host A1 (192.168.20.10 to 192.168.10.10)	PASS — 4/4 received; 0% loss

Router verification: eth1 = 192.168.10.1/24; eth2 = 192.168.20.1/24; IP forwarding = 1. The router routing table showed both connected subnets.